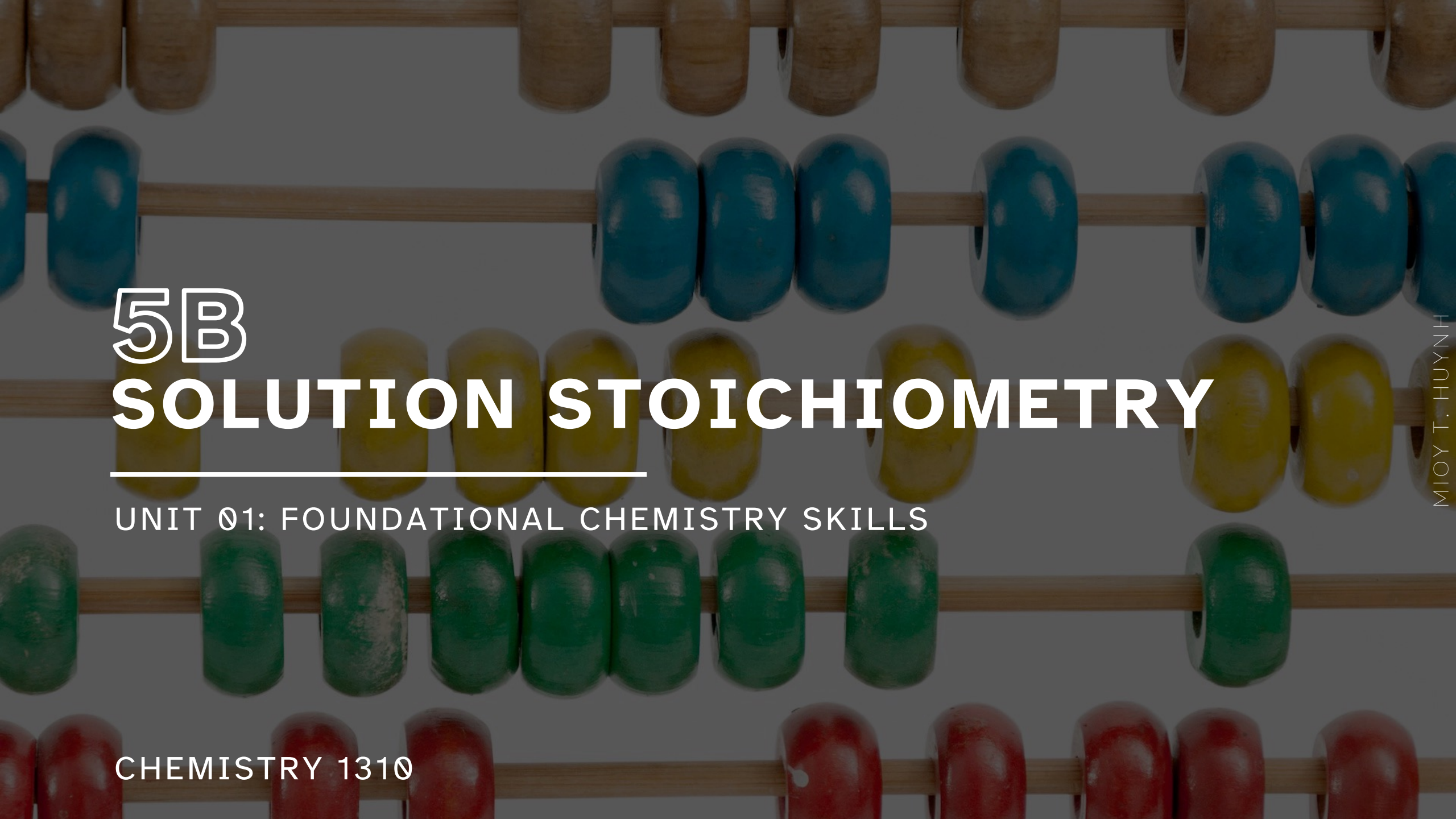




5B

SOLUTION STOICHIOMETRY

UNIT 01: FOUNDATIONAL CHEMISTRY SKILLS

CHEMISTRY 1310

LEARNING OBJECTIVES

CHAPTER 5

1. Calculate the number of moles of any substance involved in a chemical reaction from the number of moles of any other substance in the reaction.
2. Use the mass of one substance to determine the masses of other substances involved in a chemical reaction.
3. Calculate the quantities of substances produced in a reaction when quantities of more than one reactant are specified.
4. Express the quantity of product obtained from a reaction as a percentage of what the reaction is theoretically capable of producing.
5. Solve problems using molarity.
6. Extend the concept of molarity to the individual ions in solutions of ionic substances.

Theoretical Yield and Percent Yield

5.4

Theoretical yield

- The maximum amount of product that a reaction can possibly make.

Actual yield

- The amount the reaction produces in the laboratory.

Percent yield

- The ratio of actual yield to theoretical yield as a percentage.

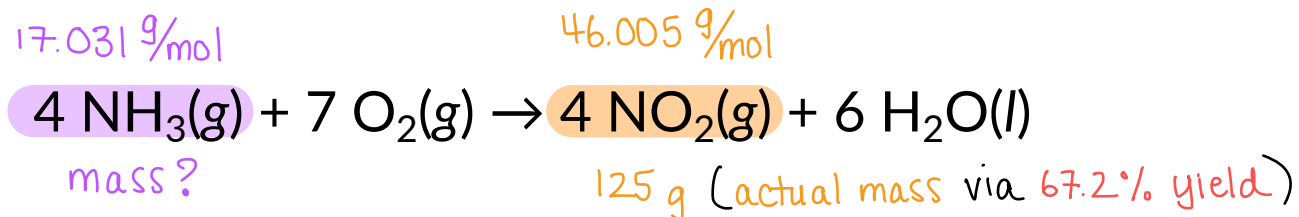
$$\% \text{ yield} = \frac{\text{actual}}{\text{theoretical}} \times 100\%$$

note: can use mass or moles;
should be $< 100\%$.

IN-CLASS QUESTION 1

Learning Objective(s): 5.4

What mass of NH_3 is required to produce 125 g of NO_2 if the percent yield for the reaction is 67.2 %?



Q1: What was expected theoretical mass of NO_2 ?

$$\% \text{ yield} = \frac{\text{actual}}{\text{theoretical}} \times 100\% \longrightarrow 67.2\% = \frac{125 \text{ g}}{\text{theoretical}} \times 100\% \longrightarrow \text{theoretical} = \frac{100}{67.2} \times 125 \text{ g} = 186. \text{ g NO}_2 \quad \begin{array}{l} (100\% \text{ yield}) \\ \text{theoretical mass} \end{array}$$

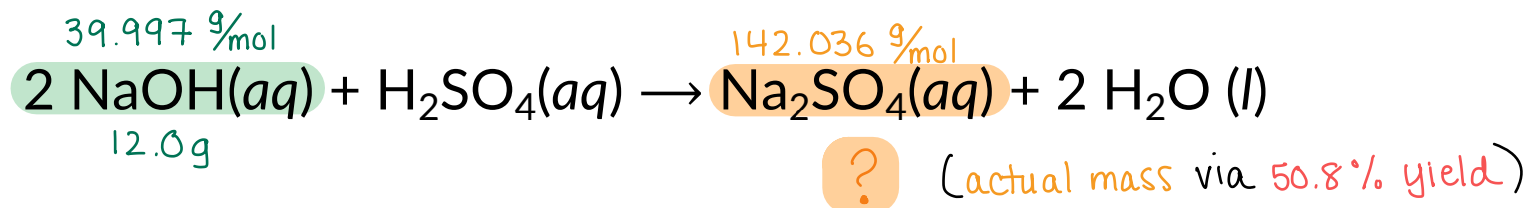
Q2: Since yield is only 67.2 %, you run reaction expecting theoretical mass. How much NH_3 to get 186. g NO_2 ?

$$186. \text{ g NO}_2 \times \frac{1 \text{ mol NO}_2}{46.005 \text{ g NO}_2} \times \frac{4 \text{ mol NH}_3}{4 \text{ mol NO}_2} \times \frac{17.031 \text{ g NH}_3}{1 \text{ mol NH}_3} = 68.9 \text{ g NH}_3$$

EXAMPLE PROBLEM

Learning Objective(s): 5.4

The reaction below has a yield of 50.8 %. If you start with 12.0 g NaOH and excess H₂SO₄, then what is the actual yield (in grams) of Na₂SO₄?



Q1: What was expected theoretical mass of Na₂SO₄?

$$12.0 \text{ g NaOH} \times \frac{1 \text{ mol NaOH}}{39.997 \text{ g NaOH}} \times \frac{1 \text{ mol Na}_2\text{SO}_4}{2 \text{ mol NaOH}} \times \frac{142.036 \text{ g Na}_2\text{SO}_4}{1 \text{ mol Na}_2\text{SO}_4} = 21.31 \text{ g Na}_2\text{SO}_4$$

(100% yield)
theoretical mass

Q2: Since yield is only 50.8 %, what is actual mass of Na₂SO₄?

$$\% \text{ yield} = \frac{\text{actual}}{\text{theoretical}} \times 100\% \longrightarrow 50.8\% = \frac{\text{actual}}{21.31 \text{ g}} \times 100\% \longrightarrow \text{actual} = 10.8 \text{ g Na}_2\text{SO}_4$$

(50.8% yield)
actual mass

Definition and Uses of Molarity

5.5

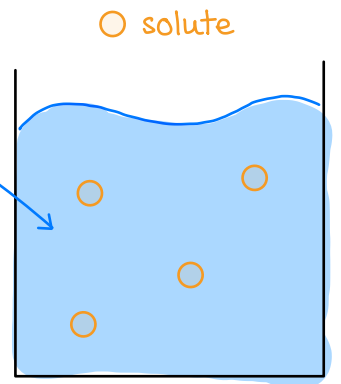
Aqueous solutions

- Result when a substance is dissolved in water.

Components of a solution:

- **Solvent:** Component present in greater amounts (usually much greater amounts); the major component
 - Water is the solvent in aqueous solutions.
- **Solute:** Component present in lesser amounts; the minor component

In all solutions, the solute dissolves in the solvent.



Definition and Uses of Molarity

5.5

Concentration

- Measure of amount of solute dissolved in the solvent
- *Concentrated*: lots of solute in solution, large value for concentration
- *Dilute*: little solute in solution, small value for concentration

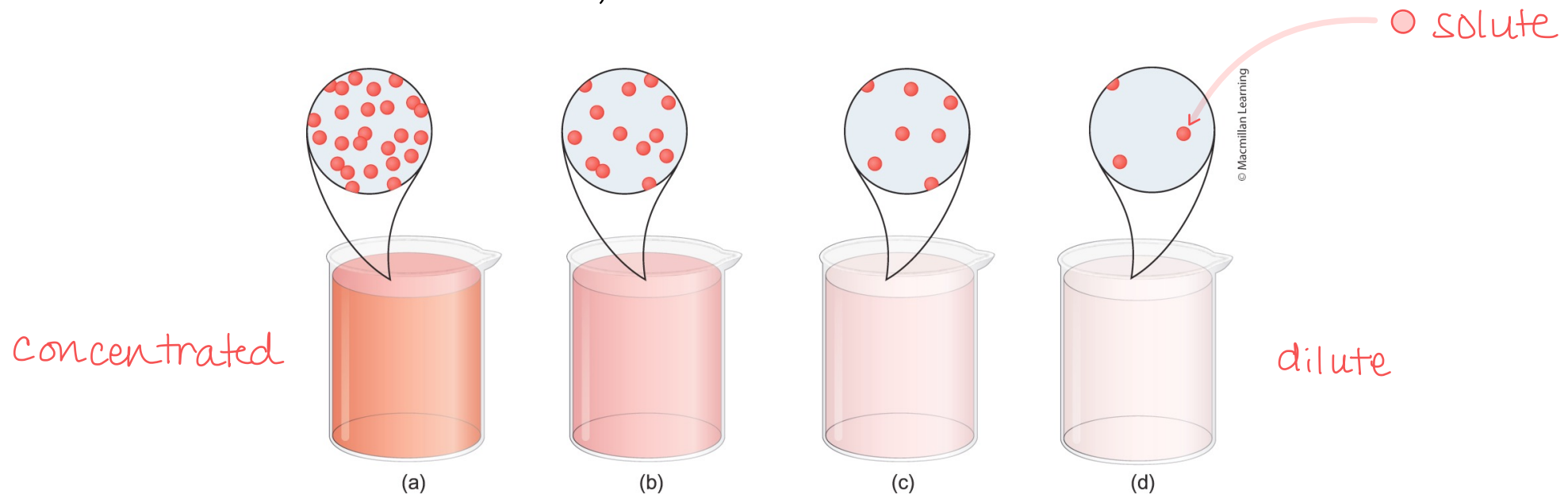


FIGURE 5.8 Microscopic depiction of (a) concentrated and (d) dilute solutions.

Definition and Uses of Molarity

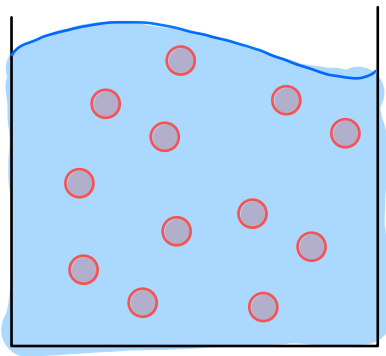
5.5

Molarity

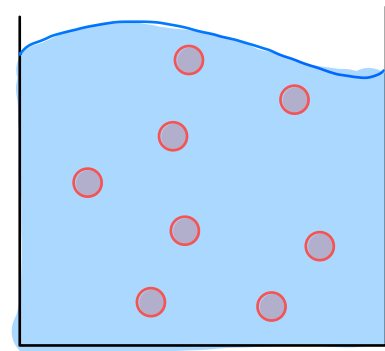
- A unit of concentration in moles of solute per liter of total solution.

$$M = \frac{\text{moles solute } \circ}{\text{volume (L) solution}} \quad ; \quad \frac{\text{mol}}{\text{L}} = M$$

* note: square brackets denote molarity, $[\]$



both 4 L total solution



$$M = \frac{12 \text{ moles solute}}{4 \text{ L solution}} = 3 \text{ M}$$

$$M = \frac{8 \text{ moles solute}}{4 \text{ L solution}} = 2 \text{ M}$$

Definition and Uses of Molarity

5.5

Dilution

- Lowering the concentration of a solution by *adding more solvent*.
- The number of moles of solute does not change during a dilution.

* no change in quantity of solute , moles

note: $M \cdot V = \# \text{ mol}$

$$M_1 V_1 = M_2 V_2$$

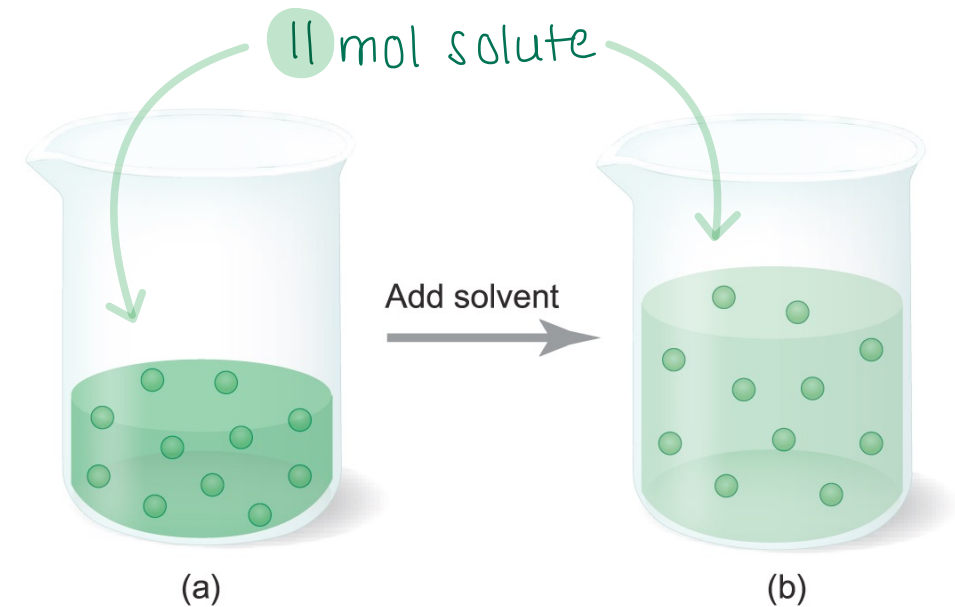


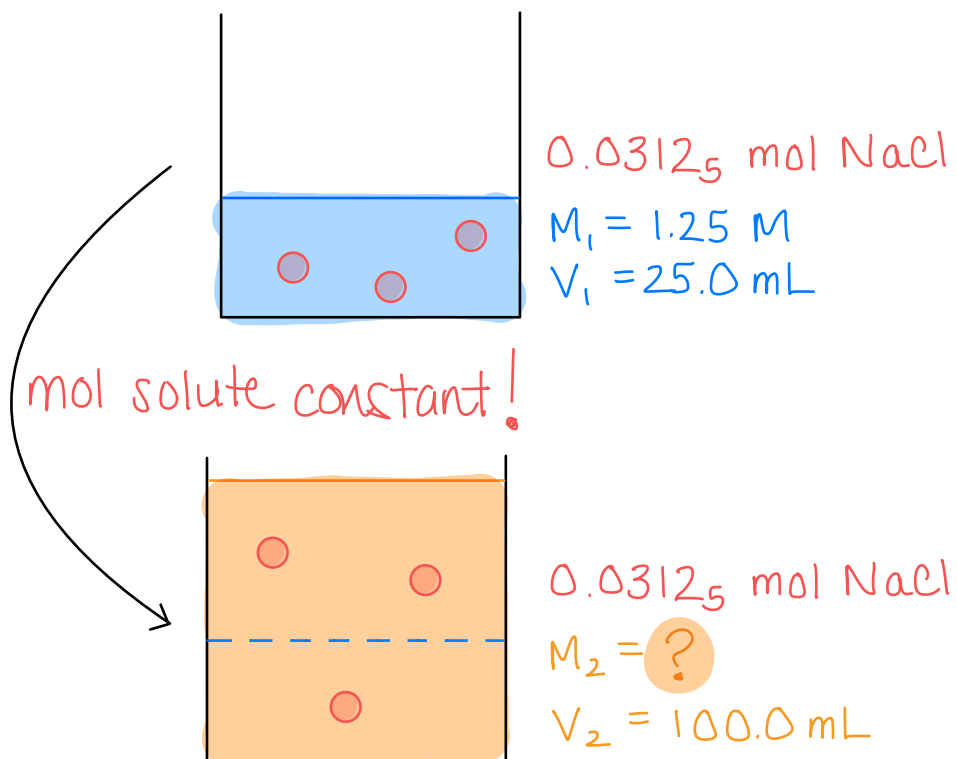
FIGURE 5.11

IN-CLASS QUESTION 2

Learning Objective(s): 5.5

A 1.25 M solution of ^{solute} NaCl is diluted from 25.0 mL to 100.0 mL.

What is the concentration (in M) of the diluted NaCl solution?



$$M_1 V_1 = M_2 V_2$$
$$\frac{1.25 \text{ mol NaCl}}{1 \text{ L}} \times 0.0250 \text{ L} = M_2 \times 0.1000 \text{ L}$$

$$0.03125 \text{ mol NaCl} = M_2 \times 0.1000 \text{ L}$$

$$\frac{0.03125 \text{ mol NaCl}}{0.1000 \text{ L}} = M_2$$

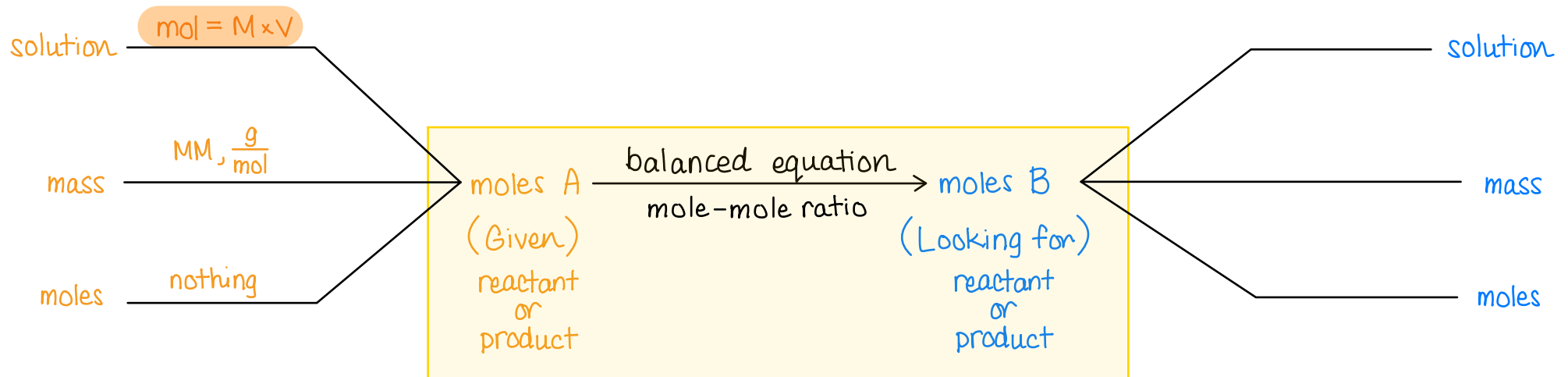
$$0.313 \text{ M} = M_2$$

Definition and Uses of Molarity

5.5

Solution stoichiometry

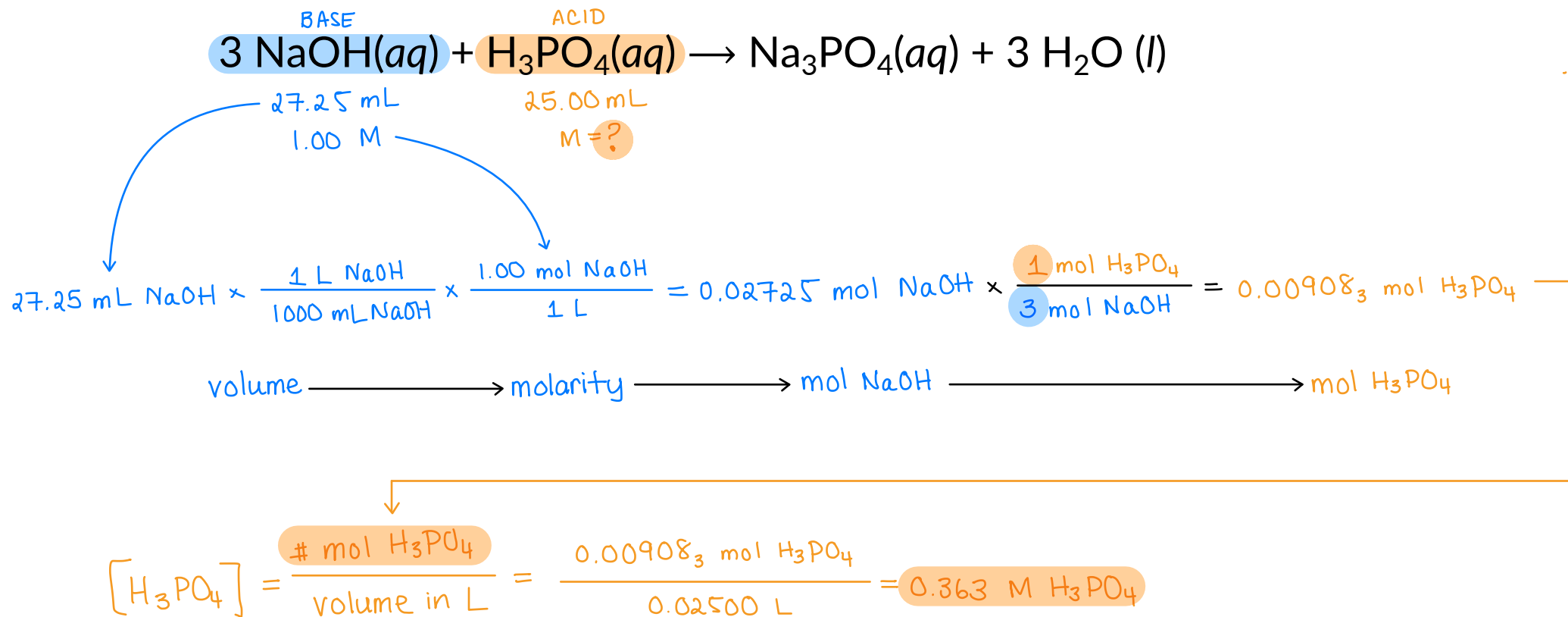
- Reactants and products may be given as molarities.
- Molarities and volumes can be used to calculate the number of moles.



EXAMPLE PROBLEM

Learning Objective(s): 5.5

It takes 27.25 mL of 1.00 M NaOH to completely react with 25.00 mL of H_3PO_4 of unknown concentration. What is the initial concentration (in M) of the acid solution?



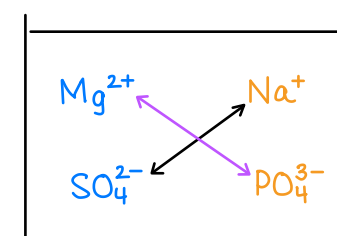
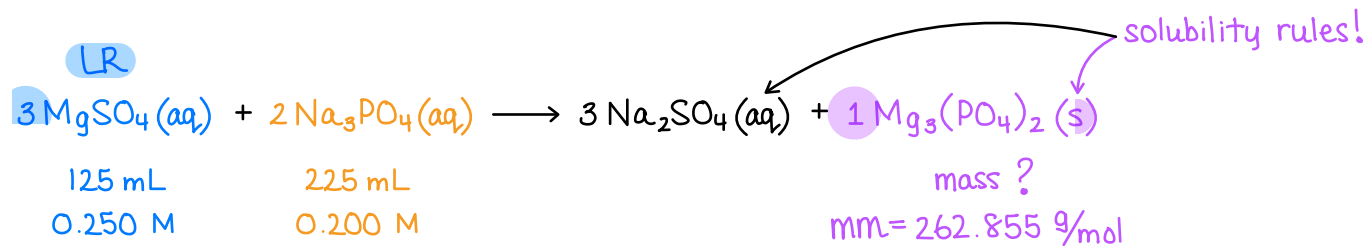
IN-CLASS QUESTION 3

Learning Objective(s): 5.5

TIP This is a LR problem because quantities of ALL reactants provided (i.e. they're limited)

What mass (in grams) of magnesium phosphate forms when 125 mL of 0.250 M MgSO_4 and 225 mL of 0.200 M Na_3PO_4 are mixed? You may assume 100 % yield.

Write chemical equation.



Identify limiting reactant (LR).

$$0.125 \text{ L MgSO}_4 \times \frac{0.250 \text{ mol MgSO}_4}{1 \text{ L}} \times \frac{1 \text{ mol Mg}_3(\text{PO}_4)_2}{3 \text{ mol MgSO}_4} = 0.01041 \text{ mol Mg}_3(\text{PO}_4)_2 \longrightarrow \text{MgSO}_4 \text{ is LR!}$$

$$0.225 \text{ L Na}_3\text{PO}_4 \times \frac{0.200 \text{ mol Na}_3\text{PO}_4}{1 \text{ L}} \times \frac{1 \text{ mol Mg}_3(\text{PO}_4)_2}{2 \text{ mol Na}_3\text{PO}_4} = 0.0225 \text{ mol Mg}_3(\text{PO}_4)_2$$

Determine mass $\text{Mg}_3(\text{PO}_4)_2$ from LR and stoichiometry.

$$0.01041 \text{ mol Mg}_3(\text{PO}_4)_2 \times \frac{262.855 \text{ g Mg}_3(\text{PO}_4)_2}{1 \text{ mol Mg}_3(\text{PO}_4)_2} = 2.74 \text{ g Mg}_3(\text{PO}_4)_2$$

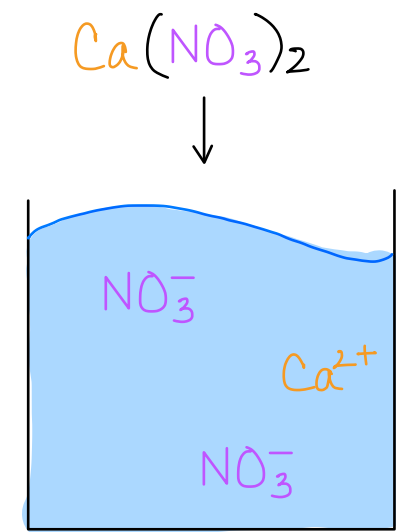
Molarities of Ions

5.6

Strong electrolytes dissociate into ions when dissolved in water.

- **Molarity of any ion:** the number of moles of that ion per liter of solution
- The subscripts in a chemical formula indicate the mole ratios of the ionic compound to its constituent elements/ions.

→ Soluble ionic compounds, strong acids, strong bases



IN-CLASS QUESTION 4

Learning Objective(s): 5.5, 5.6

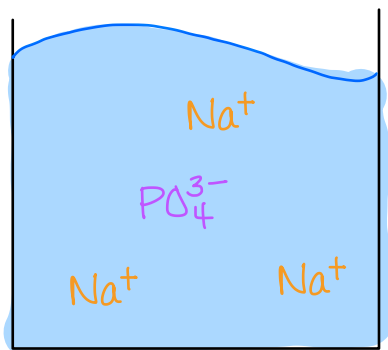
A solution is made by dissolving 20.5 g of Na_3PO_4 {163.94 g/mol} in enough water to produce 500.0 mL of solution.

Determine the following concentrations (in molarity, M) for the solution.

$$[\text{Na}_3\text{PO}_4] = \underline{0.250 \text{ M}}$$

$$[\text{Na}^+] = \underline{0.750 \text{ M}}$$

$$[\text{PO}_4^{3-}] = \underline{0.250 \text{ M}}$$



$$20.5 \text{ g Na}_3\text{PO}_4 \times \frac{1 \text{ mol Na}_3\text{PO}_4}{163.94 \text{ g Na}_3\text{PO}_4} = 0.125 \text{ mol Na}_3\text{PO}_4 / 0.500 \text{ L} = 0.250 \text{ M Na}_3\text{PO}_4$$

$$0.125 \text{ mol Na}_3\text{PO}_4 \times \frac{3 \text{ mol Na}^+}{1 \text{ mol Na}_3\text{PO}_4} = 0.375 \text{ mol Na}^+ / 0.500 \text{ L} = 0.750 \text{ M Na}^+$$

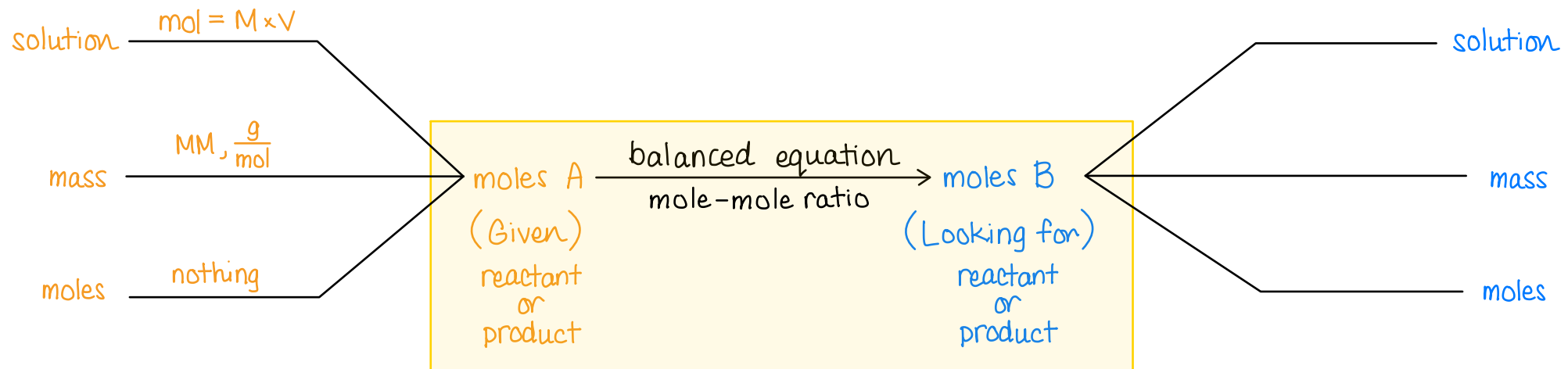
$$0.125 \text{ mol Na}_3\text{PO}_4 \times \frac{1 \text{ mol PO}_4^{3-}}{1 \text{ mol Na}_3\text{PO}_4} = 0.125 \text{ mol PO}_4^{3-} / 0.500 \text{ L} = 0.250 \text{ M PO}_4^{3-}$$

Calculations Involving Other Quantities

5.7

There are many ways to convert to moles:

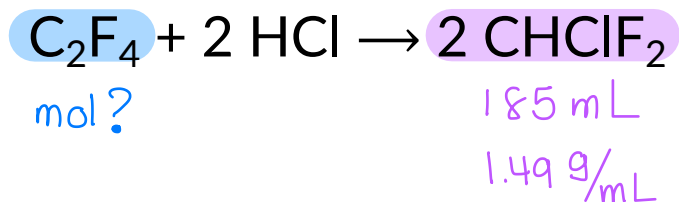
- Number of particles to moles via Avogadro's number
- Mass to moles via molar mass
- Volume to moles via molarity of solution
- Volume to mass to moles via density and molar mass of a pure substance



EXAMPLE PROBLEM

Learning Objective(s): 5.5, 5.6

Calculate the number of moles of C_2F_4 needed to produce 185 mL of CHClF_2 . The density of CHClF_2 is 1.49 g/mL.



$$\begin{array}{ccccccc} \text{volume} & & \text{density} & & \text{molar mass} & & \text{mole ratio} \\ 185 \text{ mL CHClF}_2 & \times & \frac{1.49 \text{ g CHClF}_2}{1 \text{ mL CHClF}_2} & \times & \frac{1 \text{ mol CHClF}_2}{86.465 \text{ g CHClF}_2} & \times & \frac{1 \text{ mol C}_2\text{F}_4}{2 \text{ mol CHClF}_2} = 1.59 \text{ mol C}_2\text{F}_4 \end{array}$$

BONUS QUESTION

Assessed on: Fall 2021, Exam 1

What mass (in grams) of HCl is required to produce 16.66 g of AlCl_3 if the percent yield for the following reaction is 57.15 %?

